IMPLEMENTATION OF CLASSFUL AND CLASSLESS ADDRESSING USING CISCO PACKET TRACER

ASCII Conversion

First letter of my name: **M**
Uppercase ASCII value of M: **77**

Therefore, all IP addresses in the network start with:

77.x.x.x

Classful Addressing Implementation

Since 77 lies between 1–126, it belongs to **Class A**.

Default Class A network range:

1.0.0.0 – 126.255.255.255

Network Configuration (Classful)

LAN Network:

Network: 77.0.0.0/24
Router LAN IP: 77.0.0.1
Subnet Mask: 255.255.255.0
DHCP: Disabled

WAN Network:

Router WAN IP: 77.1.0.10
Subnet Mask: 255.255.255.0

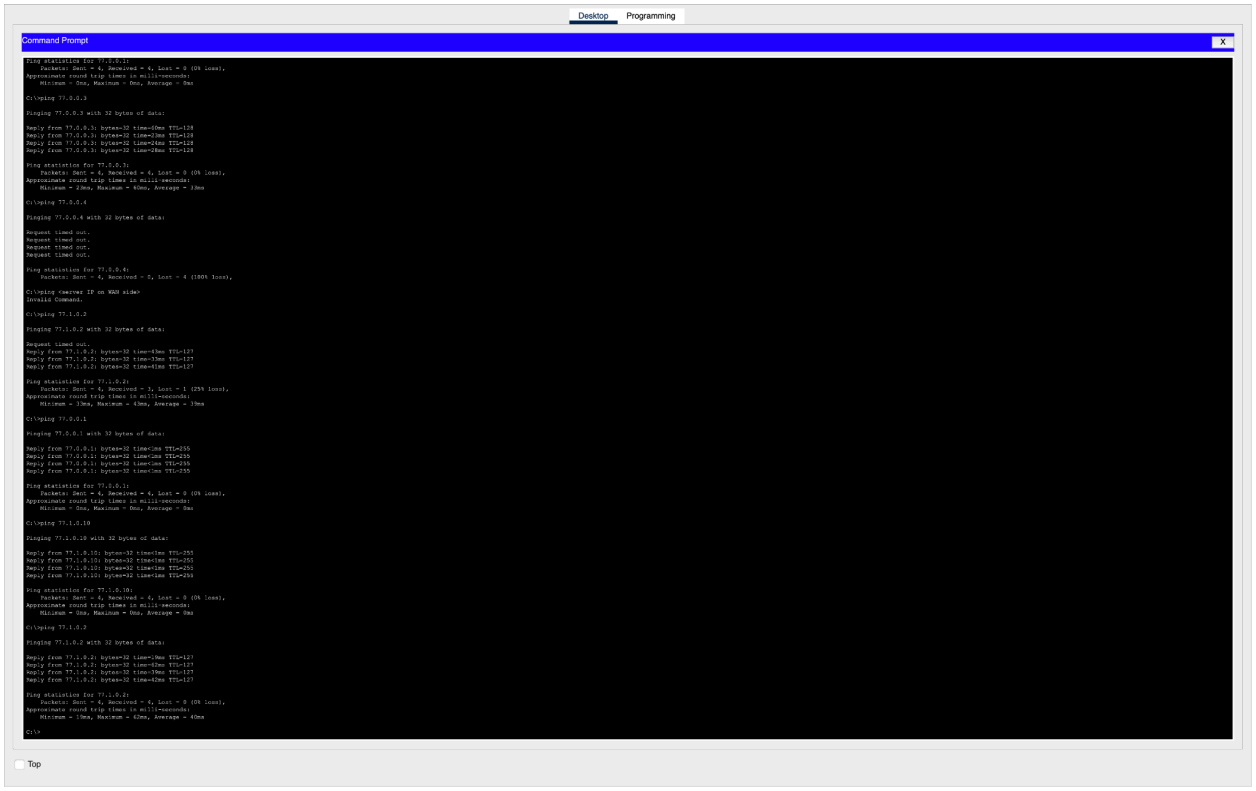
Device IP Addressing:

Device	IP Address	Subnet Mask	Default Gateway
PC	77.0.0.2	255.255.255.0	77.0.0.1

Laptop	77.0.0.3	255.255.255.0	77.0.0.1
Server	77.1.0.2	255.255.255.0	77.1.0.10

Verification

Connectivity was tested using the `ping` command.
Successful replies were received, confirming proper routing and addressing.



Classless Addressing Implementation

The classful file was duplicated and modified to implement **classless addressing using subnetting**.

In this configuration, custom subnet masks were used instead of default class-based addressing.

Subnet Used

Subnet Mask: 255.255.255.192

CIDR Notation: /26

This divides the network into smaller subnets of 64 IP addresses each.

Network Configuration (Classless)

LAN Network

Network: 77.10.10.0/26

Router LAN IP: 77.10.10.1

Subnet Mask: 255.255.255.192

DHCP: Disabled

WAN Network

Network: 77.20.20.0/26

Router WAN IP: 77.20.20.1

Subnet Mask: 255.255.255.192

Device IP Addressing (Classless)

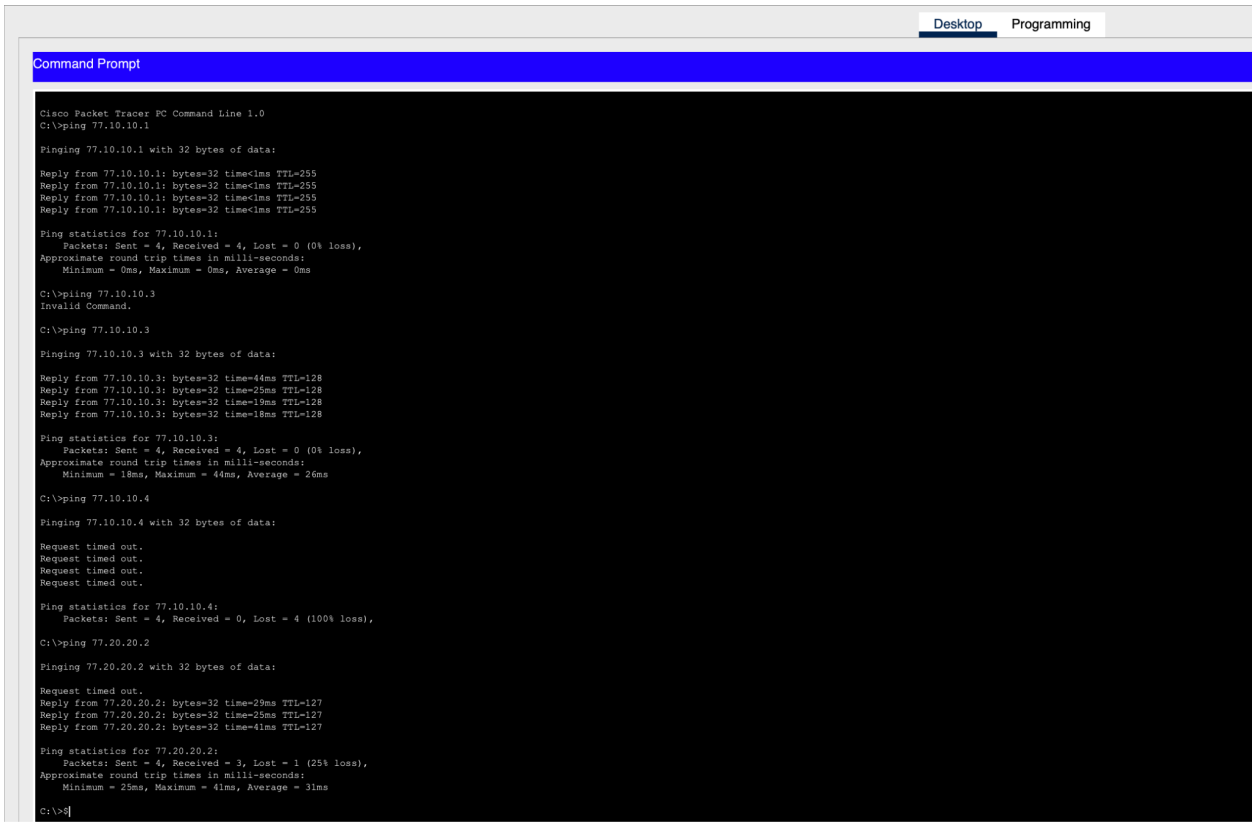
LAN Devices

Device	IP Address	Subnet Mask	Default Gateway
PC	77.10.10.2	255.255.255.192	77.10.10.1
Laptop	77.10.10.3	255.255.255.192	77.10.10.1

WAN Device

Device	IP Address	Subnet Mask	Default Gateway
Server	77.20.20.2	255.255.255.192	77.20.20.1

Verification

Connectivity was verified using the `ping` command from the PC.
The server responded successfully, confirming that:

- Routing between LAN and WAN is functioning properly.
- Subnetting has been correctly implemented.
- Classless addressing configuration is successful.